

From Curiosity to Understanding



Draw or show an object.



Explore its structure.



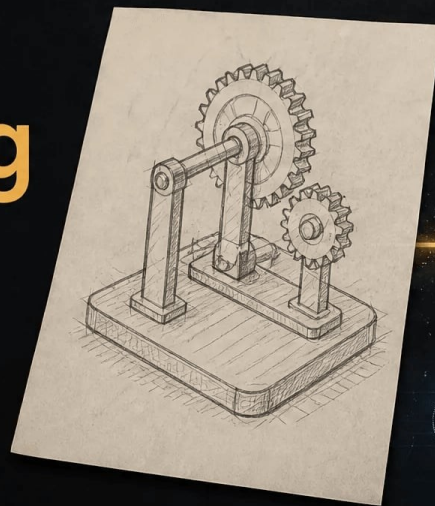
AIO 2026



Intelligent
Education



Team Iris



Object Structure

- Base
- Support
- Axle
- Large Gear
- Small Gear

Properties

Gears	2
Connections	3
Degrees of Freedom	1

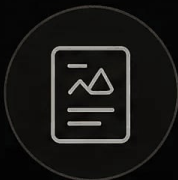


From Physical Sketches to Interactive 3D Exploration Using Computer Vision

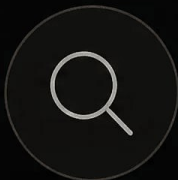


OBSERVER

CURIOSITY → FRICTION



STATIC CONTENT



SEARCH



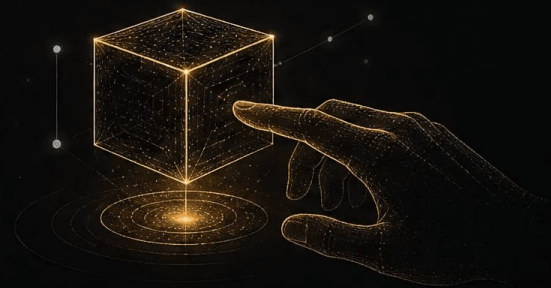
MENUS

THE GAP



EXPLORER

CURIOSITY → INTERACTION



THE GOAL:
DIRECT INTERACTION

Students are naturally **curious**.

But there is **a gap** between asking “*What is inside this?*”
and being able to **interact** with the answer.

INSIDEOUT

EXPLORE / Pinch / grab

Input: Camera 1

Camera on

Library

Parts

Debug - D

Capture: Space

Open drawing

Auto-capture stable paper

☐

Reduce motion

New drawing - N



Human heart

Human heart / 14 parts

right cardiac atrium

Explode



Depth

1

- +

Isolate - I

Hide/show - H

Show all

Reassemble - R

Controls

SLIDE 3 WHAT IF CURIOSITY WAS THE INTERFACE?

INSIDEOUT

Draw what you're
curious about.
Then explore it
with your hands.



DRAW

Sketch on paper



SEE

We capture it



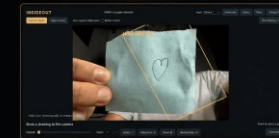
3D

We reconstruct it



EXPLORE

We explore it with your hands



InsideOut turns curiosity into tangible understanding.

LIVE DEMO

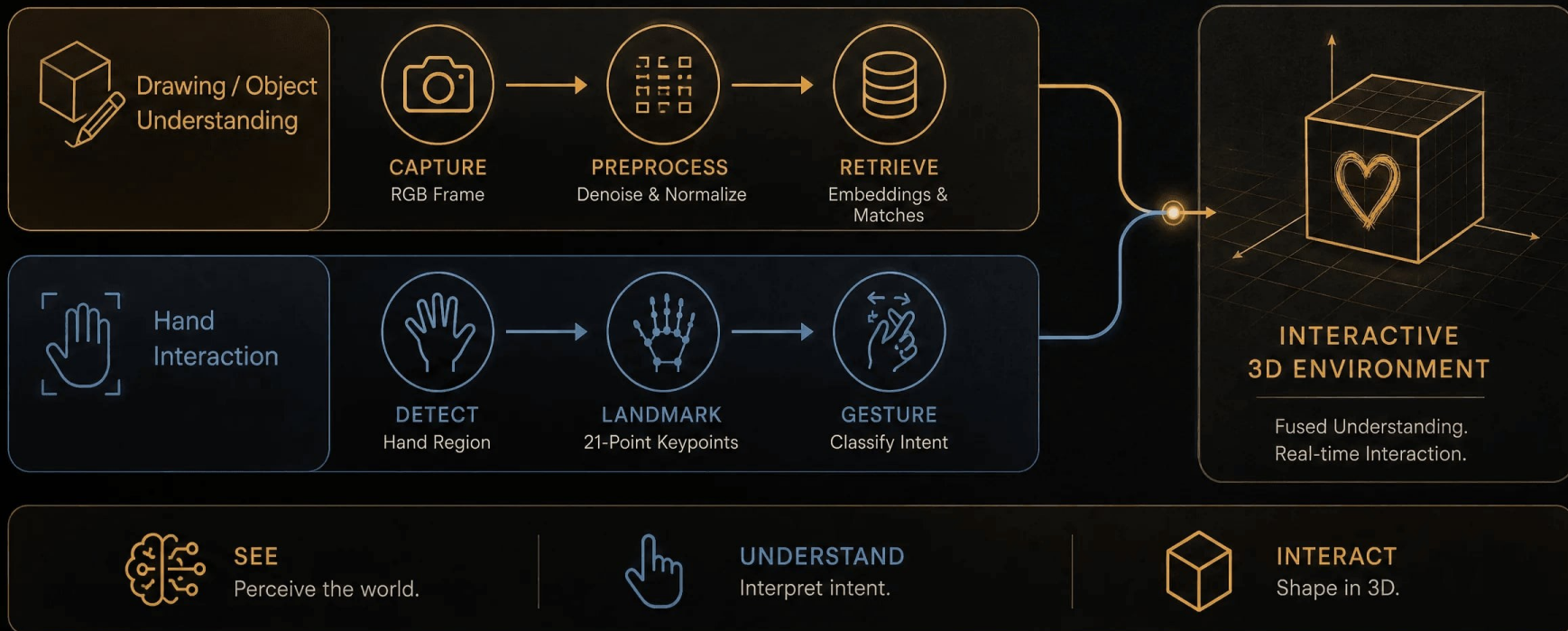
Let's try it.

Let's see the system in action.



HOW INSIDEOUT SEES.

Technical architecture simplified for judges.



YOUR HANDS BECOME THE CONTROLLER.

No controller.

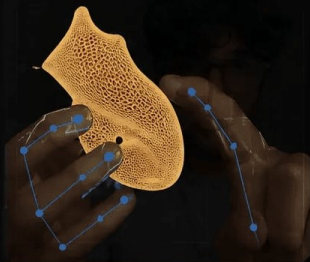
No VR headset.

Just a standard camera
and your hands.



① PINCH

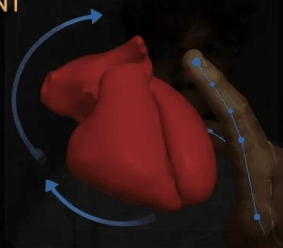
Grab to select



②

HAND MOVEMENT

Rotate to explore



③

TWO HANDS SPREAD

Spread to explode



THE INNOVATION IS THE INTERACTION CHAIN.

Defensible **innovation** claim.

EXISTING: ISOLATED TECHNOLOGIES



3D



Sketch
retrieval

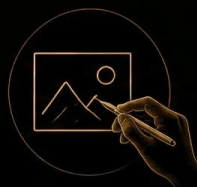


Hand
tracking

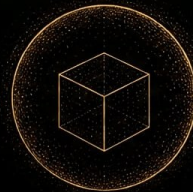
Disconnected. Manual. Cognitive overhead.

VS

INSIDEOUT: INTEGRATED WORKFLOW



DRAW
Express idea



UNDERSTAND
AI-powered
interpretation



INTERACT
Hands in
3D space

Connected. Seamless. Natural.



We connected the learner's drawing and the learner's hands
into **one continuous exploration experience.**

EDUCATION STORY

Some concepts are easier to **question** when you can actually **explore** them.

SPATIAL LEARNING CHALLENGES



DEPTH

Hard to judge what's in front, behind, or inside.



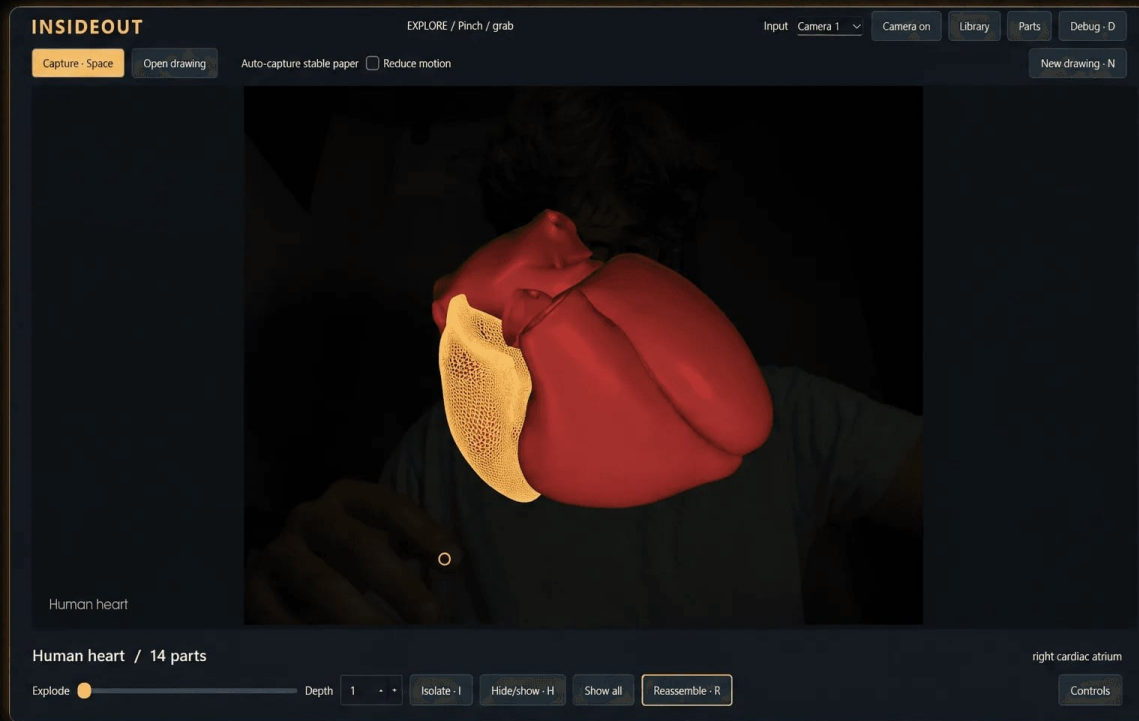
INTERNAL STRUCTURE

Complex parts are hidden and hard to imagine.



ORIENTATION

Difficult to keep track of angles and perspectives.



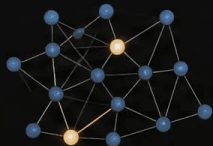
BUILT TODAY / NEXT

From a working MVP to real-world impact.

✓ BUILT TODAY (MVP)



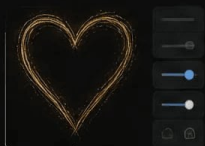
Drawing Input



CV Retrieval



Curated Library



Interactive Display



Hand Tracking



Part Exploration



○ NEXT (FUTURE)



Larger Library

More diverse shapes and styles.



More Subjects

Broader demographics and users.



Mobile Support

Take the experience anywhere.



Educational Studies

Evaluate learning outcomes and impact.



CREDIBILITY MEANS KNOWING THE LIMITS.

We are transparent about current constraints to build trust and improve responsibly.



CURATED LIBRARY

Quality depends on the diversity and representativeness of our reference collection.

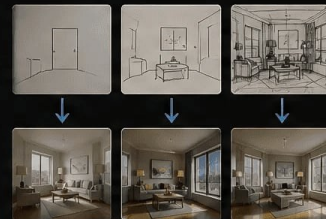


Reference Library



SKETCH VARIABILITY

Output quality varies with sketch clarity, style, and level of detail.



Low Detail → High Detail



LIGHTING SENSITIVITY

Results are sensitive to lighting conditions and exposure in input images.



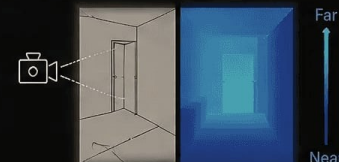
Underexposed

Overexposed



MONOCULAR DEPTH AMBIGUITY

Single-view inputs cannot fully resolve depth; scale and occlusions remain ambiguous.



NO EDUCATIONAL OUTCOME CLAIMS YET.

We do not claim learning improvement or educational benefits at this stage.